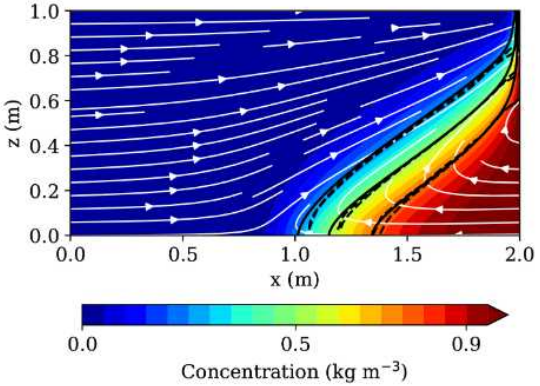
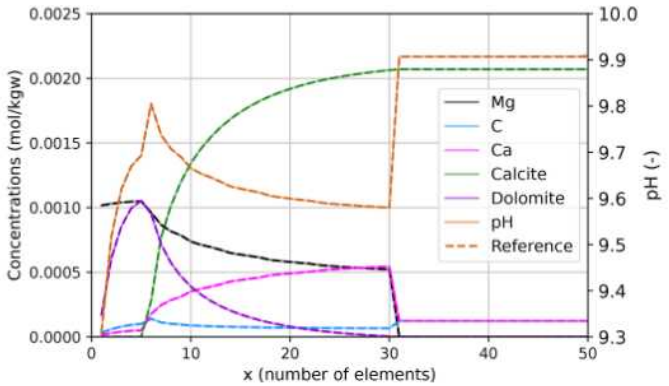
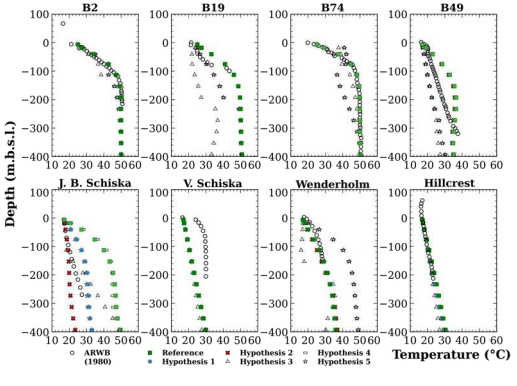
TRANSPORT Simulation Environment

Installation

Thomas Kempka
GFZ German Research Centre for Geosciences
University of Potsdam







Software prerequisites: mambaforge and git

https://github.com/conda-forge/miniforge#mambaforge

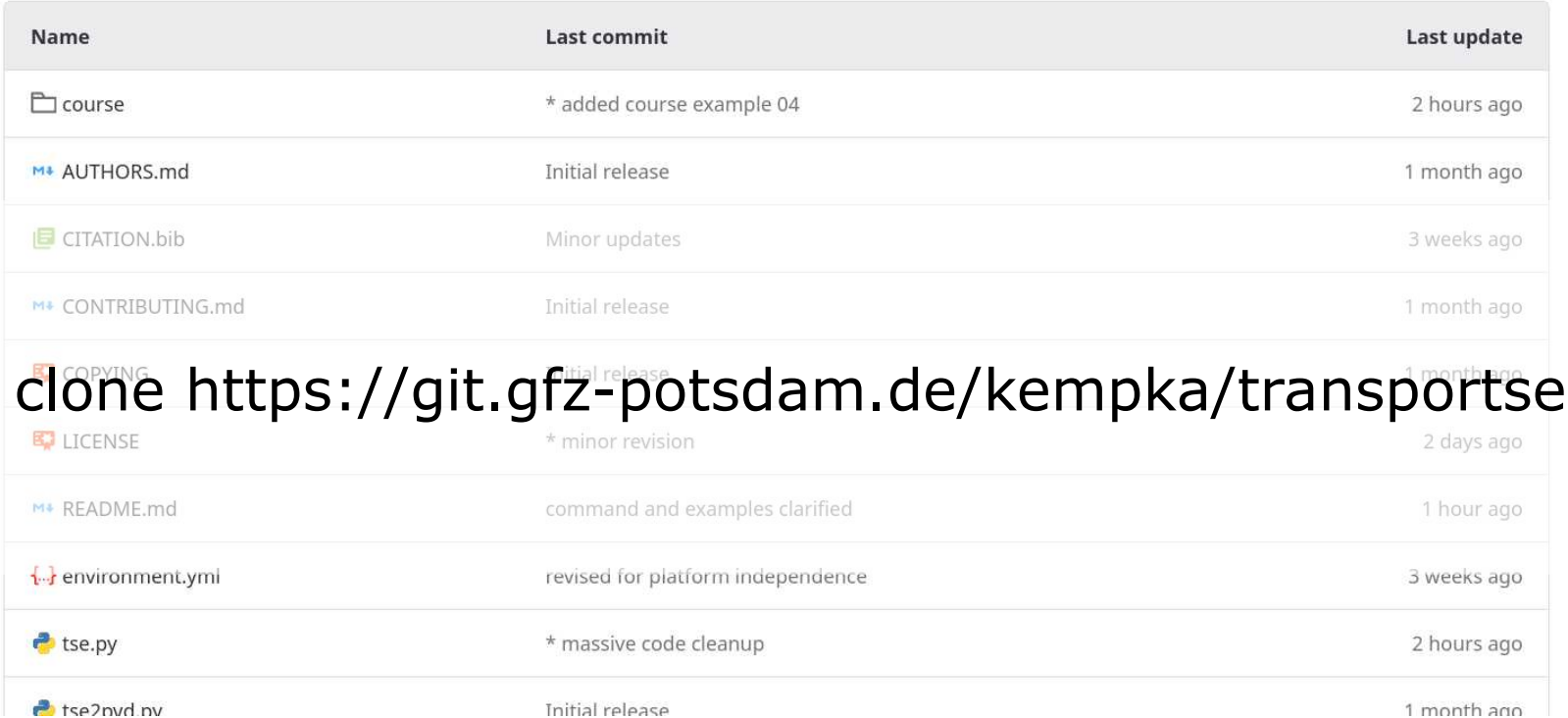
Mambaforge

Latest installers with Mamba in the base environment:

OS	Architecture	Download
Linux	x86_64 (amd64)	Mambaforge-Linux-x86_64
Linux	aarch64 (arm64)	Mambaforge-Linux-aarch64
Linux	ppc64le (POWER8/9)	Mambaforge-Linux-ppc64le
OS X	x86_64	Mambaforge-MacOSX-x86_64
OS X	arm64 (Apple Silicon)	Mambaforge-MacOSX-arm64
Windows	x86_64	Mambaforge-Windows-x86_64

```
git: mamba|conda install git
```

Clone TRANSPORTSE git repository



Name	Last commit	Last update
course	* added course example 04	2 hours ago
AUTHORS.md	Initial release	1 month ago
CITATION.bib	Minor updates	3 weeks ago
CONTRIBUTING.md	Initial release	1 month ago
COPYING	Initial release	1 month ago
LICENSE	* minor revision	2 days ago
README.md	command and examples clarified	1 hour ago
environment.yml	revised for platform independence	3 weeks ago
tse.py	* massive code cleanup	2 hours ago
tse2pvd.py	Initial release	1 month ago

git clone https://git.gfz-potsdam.de/kempka/transportse.git

Install mamba/conda environment

environment.yml 207 bytes

```yaml
name: transportse
channels:
  - conda-forge
dependencies:
  - ipython
  - joblib
  - matplotlib
  - mkl
  - numba
  - numpy
  - phreeqpy
  - pip
  - pyevtk
  - scipy
  - pip:
    - icc-rt
    - intel-openmp
```

```
mamba|conda env create
  --file=environment.yml
```

Alternative: Just install most relevant python libraries into your existing environment

environment.yml 207 bytes

```yaml
name: transportse
channels:
  - conda-forge
dependencies:
  - python
  - cmcrameri
  - matplotlib
  - mkl
  - numba
  - numpy
  - phreeqpy
  - pip
  - pyevtk
  - scipy
  - pip:
    - icc-rt
    - intel-openmp
```

```
mamba|conda install joblib numba
numpy phreeqpy pyevtk ipython
```

Activate new environment

environment.yml 207 bytes

```yaml
name: transportse
channels:
  - conda-forge
dependencies:
  - ipython
  - joblib
  - matplotlib
  - mkl
  - numba
  - numpy
  - phreeqpy
  - pip
  - pyevtk
  - scipy
  - pip:
    - icc-rt
    - intel-openmp
```

```
mamba|conda activate transportse
```